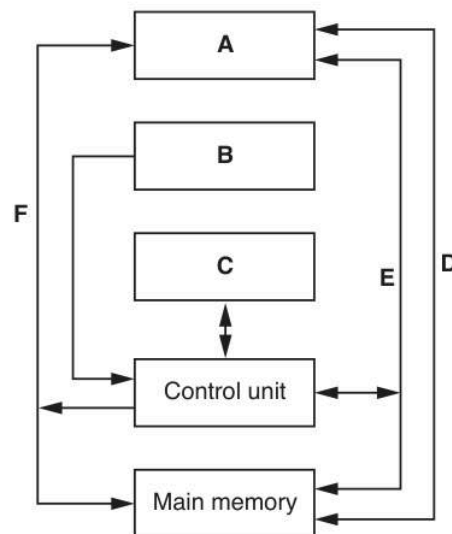


Show understanding of the basic Von Neumann model for a computer system and the stored program concept

9608_w17_qp_12_sc_4.a

QUESTION

- 4 The following diagram shows the components and buses found inside a typical personal computer (PC).



- (a) Some components and buses only have labels A to F to identify them.

For each label, choose the appropriate title from the following list. The title for label D is already given.

- Control bus
- Address bus
- Arithmetic Logic Unit (ALU)
- General purpose registers
- Secondary storage
- System clock

A

B

C

D Data bus

E

F

[5]

MARK SCHEME

4(a)	1 Mark for each correct answer A – General purpose registers B – System clock C – ALU E – Control bus F – Address bus	5
------	---	----------

9608_s15_qp_13_sc_5.a

QUESTION

5 (a) Name and describe **three** buses used in the von Neumann model.

Bus 1

Description

.....

.....

Bus 2

Description

.....

.....

Bus 3

Description

.....

.....

[6]

MARK SCHEME

5 (a) **one mark** for name of bus + **one mark** for description

address bus

- lines used to transfer address of memory or input/output location
- unidirectional bus

data bus

- used to transfer data between the processor and memory/input and output devices
- bidirectional bus

control bus

- used to transmit control signals
- e.g. read/write/fetch/ ...
- dedicated bus since all timing signals are generated according to control signal

[6]

Show understanding of the purpose and role of registers, including the difference between general purpose and special purpose registers

9608_s21_qp_12_sc_2

QUESTION

2 The following diagram shows four register notations and seven descriptions.

Draw **one** line from each register notation to its **most appropriate** description.

Register notation

MDR

CIR

MAR

PC

Description

Holds the op code and operand of an instruction ready for it to be decoded

Holds the address of the next instruction to be read

Holds flags that are set when the Arithmetic and Logic Unit (ALU) executes instructions

Holds data read from, or to be written to, memory

Holds the current value in the Index Register

Holds the address where data is to be written to or read from

Holds the result of the last instruction executed by the ALU

[4]

MARK SCHEME

2

1 mark for each register correctly described

4

Register Notation

Description

The diagram illustrates the mapping of four register notations to their respective functions. On the left, under 'Register Notation', are four boxes: MDR, CIR, MAR, and PC. On the right, under 'Description', are six boxes. Arrows indicate the following connections:

- MDR points to the first description: 'Holds the op code and operand of an instruction ready for it to be decoded'.
- CIR points to the second description: 'Holds the address of the next instruction to be read'.
- MAR points to the fifth description: 'Holds the address where data is to be written to or read from'.
- PC points to the sixth description: 'Holds the result of the last instruction executed by the ALU'.

The third and fourth descriptions, 'Holds flags that are set when the Arithmetic and Logic Unit (ALU) executes instructions' and 'Holds the current value in the Index Register', are not connected to any register notation.

Register Notation	Description
MDR	Holds the op code and operand of an instruction ready for it to be decoded
CIR	Holds the address of the next instruction to be read
MAR	Holds the address where data is to be written to or read from
PC	Holds the result of the last instruction executed by the ALU

9608_s21_qp_11_sc_3.d

QUESTION

(d) The status register contains condition flags.

Identify **three** condition flags that can be set in the status register.

- 1
- 2
- 3

[3]

MARK SCHEME

3(d)	1 mark per bullet point to max 3 e.g. • Zero • Carry • Overflow • Sign/negative • Compare results • Parity	3
------	--	----------

9608_s20_qp_11_sc_6.b

QUESTION

(b) The ACC is a general purpose register. The IX is a special purpose register.

Identify **two** other special purpose registers used in the fetch-execute cycle **and** describe their role in the cycle.

Register 1

Role

.....

.....

Register 2

Role

.....

.....

[4]

MARK SCHEME

6(b)	1 mark for correctly naming register, 1 mark for appropriate role • Program counter // PC • Stores the address of the next instruction to be fetched • Memory address register // MAR • Stores the address where data/instruction is to be read from or saved to • Memory data register // MDR • Stores data that is about to be written to memory // Stores data that has just been read from memory • Current instruction register // CIR • Stores the instruction that is currently being decoded/executed	4
------	--	----------

QUESTION

1 Von Neumann is an example of a computer architecture.

- (a) The diagram has registers used in Von Neumann architecture on the left and descriptions on the right.

Draw **one** line to match each register with its correct description.

Register	Description
Current Instruction Register	Stores the data that has just been read from memory, or is about to be written to memory
Memory Address Register	Stores the instruction that is being decoded and executed
Program Counter	Stores the address of the input device from which the processor accesses the instruction
Memory Data Register	Stores the address of the next instruction to be read
	Stores the address of the memory location about to be written to or read from

[4]

MARK SCHEME

1(a)

1 mark for each correct line

4

Register	Description
Current Instruction Register	Stores the data that has just been read from memory, or is about to be written to memory
Memory Address Register	Stores the instruction that is being decoded and executed
Program Counter	Stores the address of the input device from which the processor accesses the instruction
Memory Data Register	Stores the address of the next instruction to be read
	Stores the address of the memory location about to be written to or read from

```
graph LR; CIR[Current Instruction Register] --- D2[Stores the instruction that is being decoded and executed]; MAR[Memory Address Register] --- D4[Stores the address of the next instruction to be read]; PC[Program Counter] --- D5[Stores the address of the memory location about to be written to or read from]; MDR[Memory Data Register] --- D1[Stores the data that has just been read from memory, or is about to be written to memory];
```

9608_w18_qp_13_sc_6.a.i

QUESTION

6 The fetch-execute (FE) cycle uses special purpose registers. (a) The stages in the FE cycle are shown in register transfer notation. MAR ← [.....] PC ← PC + 1 ← [[MAR]] ← [MDR] (i) The steps shown in part (a) are incomplete. Write the missing register names in the spaces in part (a).	[3]
--	-----

MARK SCHEME

6(a)(i)	MAR ← [PC] PC ← PC + 1 MDR ← [[MAR]] CIR ← [MDR]	3
---------	---	---

QUESTION

(b) Describe the role of the Status Register and Program Counter (PC).

Status Register

.....

.....

.....

PC

.....

.....

.....

[4]

MARK SCHEME

3(b)	1 mark per bullet to max 2 for each group Status Register is interpreted as independent bits / flags Each flag is set depending on an event An example: addition overflow / result of operation is zero etc. Program Counter stores the <u>address</u> of the <u>next</u> instruction to be fetched	4
------	--	----------

QUESTION

8 The Von Neumann model uses a series of registers.

(a) Explain what is meant by the term **register**.

.....

.....

.....

.....[2]

MARK SCHEME

8(a)	1 mark per bullet to max 2 Small piece / word of (fast) memory Part of the processor Temporary storage of data Data is about to be / has been processed	2
------	---	----------

9608_s18_qp_11_sc_8.b.i

QUESTION

(b) (i) Explain the purpose of the Memory Data Register (MDR).

.....

.....

.....

.....[2]

MARK SCHEME

8(b)(i)	1 mark per bullet to max 2 Stores / holds data / instruction when fetched from memory Stores / holds data which is being written to memory The location accessed is the address held in the Memory Address Register (MAR)	2
---------	--	----------

9608_s18_qp_11_sc_8.b.ii

QUESTION

(ii) Name **two** registers, other than the MDR, that are used in the fetch-execute cycle.

Register 1

Register 2 [2]

MARK SCHEME

8(b)(ii)	1 mark per bullet to max 2 Current Instruction Register (CIR) Memory Address Register (MAR) Program Counter (PC) Accumulator (ACC) Index Register (IX) Status Register Interrupt Register	2
----------	--	----------

QUESTION

- 3 (a) Describe how special purpose registers are used in the fetch stage of the fetch-execute cycle.

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

- 3 (a) Four points from: [4]

- The Program Counter (PC) holds the address of the next instruction to be fetched
- The address in the Program Counter (PC) is copied to the Memory Address Register (MAR)
- The Program Counter (PC) is incremented
- The instruction is copied to the Memory Data Register (MDR)
 - from the address held in the Memory Address Register (MAR)
- The instruction from the Memory Data Register (MDR) is copied to the Current Instruction Register (CIR)

QUESTION

- 3 (a) Describe how special purpose registers are used in the fetch stage of the fetch-execute cycle.

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

- 3 (a) Four points from: [4]

- The Program Counter (PC) holds the address of the next instruction to be fetched
- The address in the Program Counter (PC) is copied to the Memory Address Register (MAR)
- The Program Counter (PC) is incremented
- The instruction is copied to the Memory Data Register (MDR)
 - from the address held in the Memory Address Register (MAR)
- The instruction from the Memory Data Register (MDR) is copied to the Current Instruction Register (CIR)

QUESTION

- (ii) Show the contents of the index register after the execution of the instruction:

DEC IX

Index register:

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

(ii) 0000 0111

[1]

QUESTION

(b) State the role of each of the following special purpose registers used in a typical processor.

Program Counter

.....

Memory Data Register

.....

Current Instruction Register

.....

Memory Address Register

.....

[4]

MARK SCHEME

(b) Program Counter – stores the address of next instruction to be executed

Memory Data Register – stores the data in transit between memory and other registers //
holds the instruction before it is passed to the CIR

Current Instruction Register – stores the current instruction being executed

Memory Address Register – stores the address of the memory location which is about to be
accessed [4]

QUESTION

(b) State the role of each of the following special purpose registers used in a typical processor.

Program Counter

.....

Memory Data Register

.....

Current Instruction Register

.....

Memory Address Register

.....

[4]

MARK SCHEME

(b) Program Counter – stores the address of next instruction to be executed

Memory Data Register – stores the data in transit between memory and other registers //
holds the instruction before it is passed to the CIR

Current Instruction Register – stores the current instruction being executed

Memory Address Register – stores the address of the memory location which is about to be
accessed

[4]

QUESTION

(c) Describe what happens to the registers when the following instruction is executed:

LDD 35

.....

.....

.....

..... [2]

MARK SCHEME

- (c) • the MAR is loaded with the operand of the instruction // loaded with 35
• the Accumulator is loaded with the contents of the address held in MAR
// the Accumulator is loaded with the contents of the address 35

[2]

Show understanding of the purpose and roles of the Arithmetic and Logic Unit (ALU), Control Unit (CU) and system clock, Immediate Access Store (IAS)

9608_w21_qp_11_sc_6.a

QUESTION

- 6 (a) Complete the following sentences that describe parts of a processor in a Von Neumann model for a computer system.

There are buses that transfer data between components in a computer system.

The width of the determines the number of directly accessible memory locations.

The sends signals on the to direct the operation of system components.

..... pulses are used to synchronise the components on the motherboard. [5]

MARK SCHEME

6(a)	One mark for each term in bold. There are three buses that transfer data between components in a computer system. The width of the address bus determines the number of directly accessible memory locations. The control unit sends signals on the control bus to direct the operation of system components. Clock pulses are used to synchronise the components on the motherboard.	5
------	--	---

QUESTION

3 A computer is designed using the Von Neumann model.

(a) Describe the role of the Arithmetic and Logic Unit (ALU) and Control Unit (CU) in the Von Neumann model.

ALU

.....

.....

.....

CU

.....

.....

.....

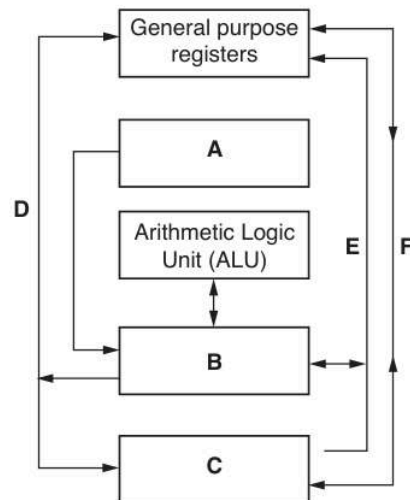
[4]

MARK SCHEME

3(a)	1 mark per bullet to max 2 for each group ALU performs arithmetic operations And logical operations / comparisons Control Unit sends / receives signals Synchronises operations to control operations // execution of instructions Accept by example e.g. Input output // flow of data	4
------	---	----------

QUESTION

- 4 (a) The diagram shows the components and buses found inside a typical Personal Computer (PC).



Some components and buses only have labels **A** to **F** to identify them.

For each label, choose the appropriate title from the following list. The title for label **D** is already given.

- Control bus
- System clock
- Data bus
- Control unit
- Main memory
- Secondary storage

A

B

C

D Address bus

E

F

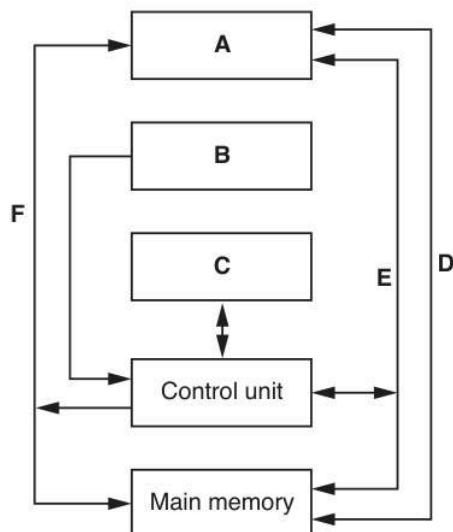
[5]

MARK SCHEME

4(a)	A – System clock B – Control unit C – Main memory E – Control bus F – Data bus	5
------	--	---

QUESTION

- 4 The following diagram shows the components and buses found inside a typical personal computer (PC).



- (a) Some components and buses only have labels **A** to **F** to identify them.

For each label, choose the appropriate title from the following list. The title for label **D** is already given.

- Control bus
- Address bus
- Arithmetic Logic Unit (ALU)
- General purpose registers
- Secondary storage
- System clock

- A**
- B**
- C**
- D** Data bus
- E**
- F**

[5]

MARK SCHEME

4(a)	1 Mark for each correct answer A – General purpose registers B – System clock C – ALU E – Control bus F – Address bus	5
------	---	----------

Show understanding of how data are transferred between various components of the computer system using the address bus, data bus and control bus

9608_w21_qp_11_sc_6.a

QUESTION

- 6 (a) Complete the following sentences that describe parts of a processor in a Von Neumann model for a computer system.

There are buses that transfer data between components in a computer system.

The width of the determines the number of directly accessible memory locations.

The sends signals on the to direct the operation of system components.

..... pulses are used to synchronise the components on the motherboard. [5]

MARK SCHEME

6(a)	One mark for each term in bold. There are three buses that transfer data between components in a computer system. The width of the address bus determines the number of directly accessible memory locations. The control unit sends signals on the control bus to direct the operation of system components. Clock pulses are used to synchronise the components on the motherboard.	5
------	--	----------

QUESTION

(b) Many components of the computer system transfer data between them using buses. One example of a bus is an address bus.

(i) Name **two** other buses that exist within a computer and give the purpose of each.

Bus 1

Purpose

.....

.....

Bus 2

Purpose

.....

.....

[4]

MARK SCHEME

1(b)(i)	1 mark for naming, 1 mark for purpose for each bus Data bus Carries data between the processor and memory / carries data that is currently being processed. Control bus Transmits signals between the control unit and the other components	4
---------	---	----------

QUESTION

(ii) State the benefit of increasing the address bus width from 16 bits to 32 bits.

.....

..... [1]

MARK SCHEME

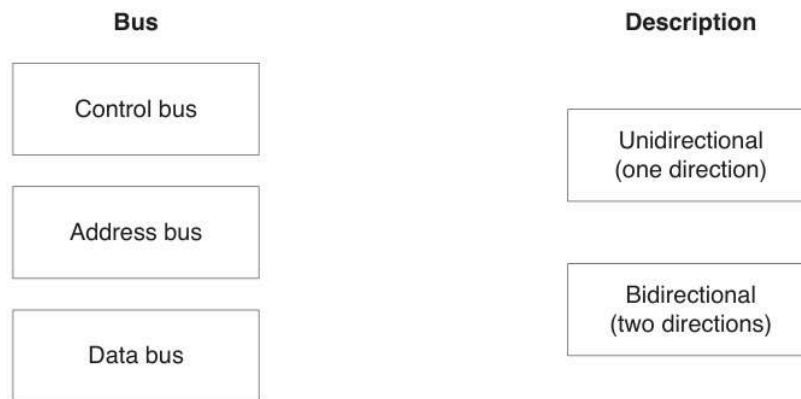
1(b)(ii)	Significant increase in the number of directly addressed memory locations // increases the number of directly addressable memory locations from 2^{16} to 2^{32}	1
----------	--	----------

QUESTION

(c) The processor uses buses in the FE cycle.

The diagram shows three buses and two descriptions.

Draw **one** line from each bus to its appropriate description.



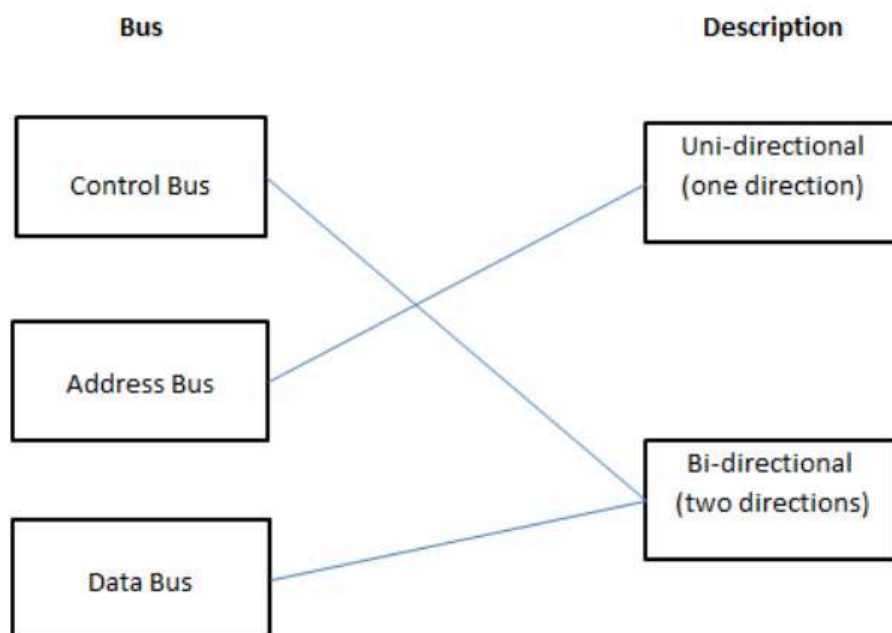
[2]

MARK SCHEME

4(c)

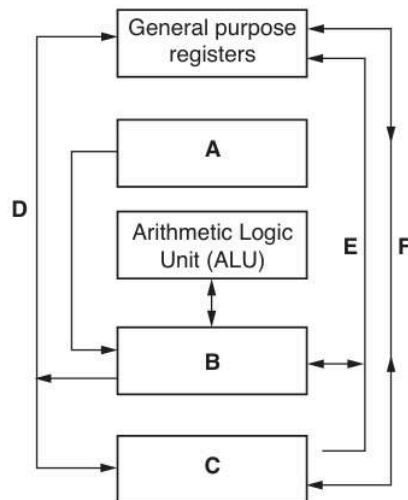
1 mark for **1 correct** connection
2 marks for all **3 correct** connections

2



QUESTION

- 4 (a) The diagram shows the components and buses found inside a typical Personal Computer (PC).



Some components and buses only have labels **A** to **F** to identify them.

For each label, choose the appropriate title from the following list. The title for label **D** is already given.

- Control bus
- System clock
- Data bus
- Control unit
- Main memory
- Secondary storage

A

B

C

D Address bus

E

F

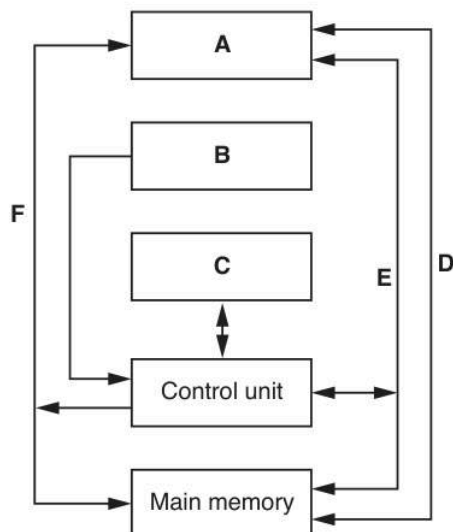
[5]

MARK SCHEME

4(a)	A – System clock B – Control unit C – Main memory E – Control bus F – Data bus	5
------	--	---

QUESTION

- 4 The following diagram shows the components and buses found inside a typical personal computer (PC).



- (a) Some components and buses only have labels **A** to **F** to identify them.

For each label, choose the appropriate title from the following list. The title for label **D** is already given.

- Control bus
- Address bus
- Arithmetic Logic Unit (ALU)
- General purpose registers
- Secondary storage
- System clock

A

B

C

D Data bus

E

F

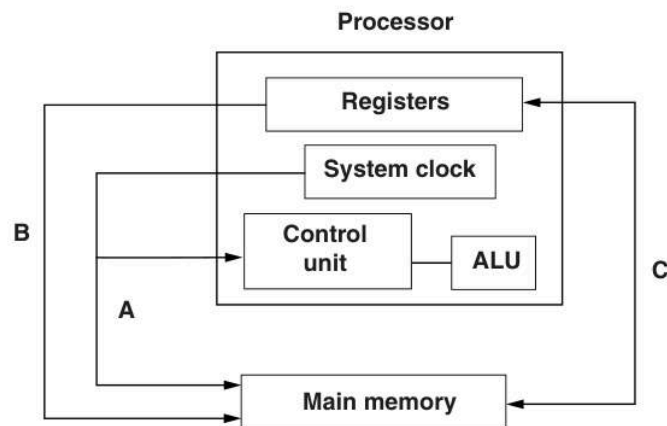
[5]

MARK SCHEME

4(a)	1 Mark for each correct answer A – General purpose registers B – System clock C – ALU E – Control bus F – Address bus	5
------	---	----------

QUESTION

2 (a)



The diagram above shows a simplified form of processor architecture.

Name the three buses labelled A, B and C.

A

B

C

[3]

MARK SCHEME

2 (a) A = control bus

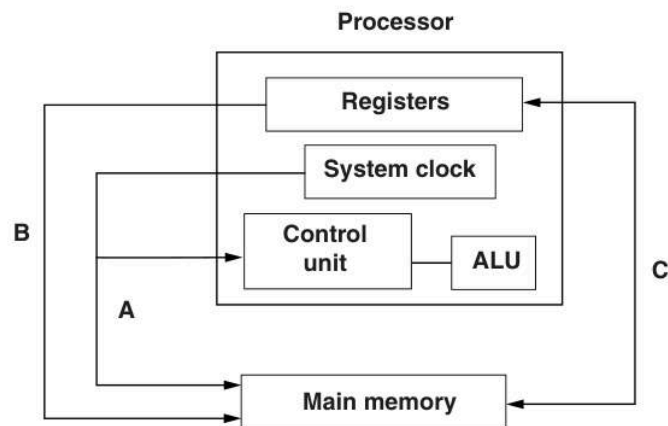
B = address bus

C = data bus

[3]

QUESTION

2 (a)



The diagram above shows a simplified form of processor architecture.

Name the three buses labelled A, B and C.

A

B

C

[3]

MARK SCHEME

2 (a) A = control bus

B = address bus

C = data bus

[3]

QUESTION

5 (a) Name and describe **three** buses used in the von Neumann model.

Bus 1

Description

.....

.....

Bus 2

Description

.....

.....

Bus 3

Description

.....

.....

[6]

MARK SCHEME

5 (a) **one mark** for name of bus + **one mark** for description

address bus

- lines used to transfer address of memory or input/output location
- unidirectional bus

data bus

- used to transfer data between the processor and memory/input and output devices
- bidirectional bus

control bus

- used to transmit control signals
- e.g. read/write/fetch/ ...
- dedicated bus since all timing signals are generated according to control signal

[6]

Show understanding of how factors contribute to the performance of the computer system

9608_w19_qp_11_sc_1.b.ii

QUESTION

(ii) State the benefit of increasing the address bus width from 16 bits to 32 bits.

.....
..... [1]

MARK SCHEME

1(b)(ii)	Significant increase in the number of directly addressed memory locations // increases the number of directly addressable memory locations from 2^{16} to 2^{32}	1
----------	--	---

9608_w17_qp_12_sc_4.b

QUESTION

(b) Clock speed is a factor that affects the performance of a PC. Explain this statement.

.....
.....
..... [2]

MARK SCHEME

4(b)	1 Mark per bullet, max 2 The clock sends out a number of pulses in a given time interval (clock speed) Each processor instruction takes a certain number of clock cycles to execute The higher the clock frequency, the shorter the execution time for the instruction // Increasing the clock frequency improves performance	2
------	--	---

QUESTION

- 8 (a) Explain how the width of the data bus and system clock speed affect the performance of a computer system.

Width of the data bus

.....

.....

.....

Clock speed

.....

.....

.....[3]

MARK SCHEME

- 8 (a) **maximum of 2 marks** for data bus width and **maximum of 2 marks** for clock speed

data bus width

- the width of the data bus determines the number of bits that can be simultaneously transferred
- increasing the width of the data bus increases the number of bits/amount of data that can be moved at one time (or equivalent)
- ...hence improving processing speed as fewer transfers are needed
- By example: e.g. double the width of the data bus moves 2x data per clock pulse

clock speed

- determines the number of cycles the CPU can execute per second
- increasing clock speed increases the number of operations/number of fetch-execute cycles that can be carried out per unit of time
- ...however, there is a limit on clock speed because the heat generated by higher clock speeds cannot be removed fast enough

[3]

QUESTION

- 8 (a) Explain how the width of the data bus and system clock speed affect the performance of a computer system.

Width of the data bus

.....

.....

.....

Clock speed

.....

.....

.....[3]

MARK SCHEME

- 8 (a) **maximum of 2 marks** for data bus width and **maximum of 2 marks** for clock speed

data bus width

- the width of the data bus determines the number of bits that can be simultaneously transferred
- increasing the width of the data bus increases the number of bits/amount of data that can be moved at one time (or equivalent)
- ...hence improving processing speed as fewer transfers are needed
- By example: e.g. double the width of the data bus moves 2x data per clock pulse

clock speed

- determines the number of cycles the CPU can execute per second
- increasing clock speed increases the number of operations/number of fetch-execute cycles that can be carried out per unit of time
- ...however, there is a limit on clock speed because the heat generated by higher clock speeds cannot be removed fast enough

[3]

Understand how different ports provide connection to peripheral devices

9608_s20_qp_13_sc_2.c.i

QUESTION

(c) Billy's computer has several ports.

(i) State the purpose of a port.

[1]

MARK SCHEME

2(c)(i)	1 mark - To provide the connection to peripheral devices - To provide an interface between the computer and other devices	1
---------	--	----------

9608_s20_qp_13_sc_2.c.ii

QUESTION

(ii) Identify **one** type of port.

[1]

MARK SCHEME

2(c)(ii)	1 mark - USB - HDMI - SCSI - Firewire - Ethernet - Any other acceptable port	1
----------	---	----------

QUESTION

(iii) The peripheral devices are plugged into USB ports of the computer.

Describe **two** benefits of connecting the peripheral devices using a USB port.

1

.....

.....

.....

2

.....

.....

.....

[4]

MARK SCHEME

2(c)(iii)	1 mark for benefit, 1 mark for expansion for max 2 benefits • Fast data transfer • ... useful when transferring large files such as video files • Automatic connection / plug-and-play • ... so usually there is no need to install separate device drivers • USB is a standard adopted by many manufacturers • ... meaning all new computers will be equipped with USB ports • Devices may be powered or charged through USB • ... so external devices may be charged while working at the computer	4
-----------	--	----------

QUESTION

(b) Most computers use Universal Serial Bus (USB) ports to allow the attachment of devices.

Describe **two** benefits of using USB ports.

1

.....

2

.....

[2]

MARK SCHEME

(b) Any **two** from:

- devices automatically detected and configured when first attached/plug and play
- it is nearly impossible to wrongly connect a device
- USB has become an industrial standard
- supported by many operating systems
- USB 3.0 allows full duplex data transfer
- later versions are backwards compatible with earlier USB systems
- allows power to be drawn to charge portable devices

[2]

9608_s15_qp_11_sc_8.b

QUESTION

(b) Most computers use Universal Serial Bus (USB) ports to allow the attachment of devices.

Describe **two** benefits of using USB ports.

1

.....

2

.....[2]

MARK SCHEME

(b) Any **two** from:

- devices automatically detected and configured when first attached/plug and play
- it is nearly impossible to wrongly connect a device
- USB has become an industrial standard
- supported by many operating systems
- USB 3.0 allows full duplex data transfer
- later versions are backwards compatible with earlier USB systems
- allows power to be drawn to charge portable devices

[2]

Describe the stages of the Fetch-Execute (F-E) cycle

9608_w21_qp_11_sc_6.b

QUESTION

(b) Describe the stages of the fetch-execute (F-E) cycle.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

MARK SCHEME

6(b)	One mark per bullet point to max 5 • The address in the program counter is the address of next item to be fetched • The address is copied into MAR ... • ...using the address bus • The instruction from that address moved/copied from main memory to MDR ... • ...using the data bus • The instruction is transferred from MDR to CIR • The processor's instruction set is used to decode the instruction// the instruction is decoded in the CIR • ...into op code and operand • The processor executes the instruction // the processor processes the data as required • The address in PC is incremented ready for next loop	5
------	---	---

QUESTION

(d) The fetch-execute (FE) cycle is shown in register transfer notation.

Complete the FE cycle using register transfer notation.

..... $\leftarrow$ [PC]

PC $\leftarrow$ + 1

MDR $\leftarrow$ [[MAR]]

..... $\leftarrow$ [MDR]

[3]

MARK SCHEME

3(d)	1 mark for each correctly completed item	3
	MAR $\leftarrow$ [PC] PC $\leftarrow$ [PC] + 1 MDR $\leftarrow$ [[MAR]] CIR $\leftarrow$ [MDR]	

QUESTION

(b) The ACC is a general purpose register. The IX is a special purpose register.

Identify **two** other special purpose registers used in the fetch-execute cycle **and** describe their role in the cycle.

Register 1

Role

.....

.....

Register 2

Role

.....

.....

[4]

MARK SCHEME

6(b)	1 mark for correctly naming register, 1 mark for appropriate role • Program counter // PC • Stores the address of the next instruction to be fetched • Memory address register // MAR • Stores the address where data/instruction is to be read from or saved to • Memory data register // MDR • Stores data that is about to be written to memory // Stores data that has just been read from memory • Current instruction register // CIR • Stores the instruction that is currently being decoded/executed	4
------	---	----------

9608_s19_qp_12_sc_3.a

QUESTION

3 The fetch-execute cycle is shown in register transfer notation.

01 $\text{MAR} \leftarrow [\text{PC}]$

02 $\text{PC} \leftarrow [\text{PC}] - 1$

03 $\text{MDR} \leftarrow [\text{MAR}]$

04 $\text{CIR} \leftarrow [\text{MAR}]$

(a) There are **three** errors in the fetch-execute cycle shown.

Identify the line number of each error and give the correction.

Line number

Correction

Line number

Correction

Line number

Correction

[3]

MARK SCHEME

3(a)	1 mark for each error and correction • Line 02 should be +1 not -1 // $\text{PC} \leftarrow [\text{PC}] + 1$ • Line 03 should be double brackets around MAR // $\text{MDR} \leftarrow [[\text{MAR}]]$ • Line 04 should be MDR not MAR // $\text{CIR} \leftarrow [\text{MDR}]$	3
------	--	----------

9608_w18_qp_13_sc_6.a.i

QUESTION

- 6 The fetch-execute (FE) cycle uses special purpose registers.
- (a) The stages in the FE cycle are shown in register transfer notation.

MAR $\leftarrow$ [.....]

PC $\leftarrow$ PC + 1

..... $\leftarrow$ [[MAR]]

..... $\leftarrow$ [MDR]

- (i) The steps shown in **part (a)** are incomplete.

Write the missing register names in the spaces in **part (a)**.

[3]

MARK SCHEME

6(a)(i)	MAR $\leftarrow$ [PC] PC $\leftarrow$ PC + 1 MDR $\leftarrow$ [[MAR]] CIR $\leftarrow$ [MDR]	3
---------	---	---

9608_w18_qp_13_sc_6.a.ii

QUESTION

- (ii) The third instruction [[MAR]] has double brackets.

State the purpose of the double brackets.

.....

.....[1]

MARK SCHEME

6(a)(ii)	1 mark from: - The contents of the MAR is an address, it is the contents of that address which is transferred to MDR - The contents of the address pointed to by the MAR is transferred to the MDR	1
----------	---	---

QUESTION

- 4 A student has written the steps of the fetch stage of the fetch-execute (FE) cycle in register transfer notation. The student has made some errors.

Line 1 $\text{MDR} \leftarrow [\text{PC}]$
 Line 2 $\text{PC} \leftarrow \text{PC} + 1$
 Line 3 $\text{MDR} \leftarrow [\text{MAR}]$
 Line 4 $\text{CIR} \leftarrow \text{PC}$

- (a) Identify the line numbers of **three** errors that the student has made. Write the correct notation for each error.

Line number of error	Correct notation

[3]

MARK SCHEME

4(a)

1 mark per correct line, max 3

3

Line number of error	Correct notation
1	MAR ← [PC]
3	MDR ← [[MAR]]
4	CIR ← [MDR]
2	PC ← [PC] + 1

QUESTION

- (ii) Name **two** registers, other than the MDR, that are used in the fetch-execute cycle.

Register 1

Register 2

[2]

MARK SCHEME

8(b)(ii)	1 mark per bullet to max 2 Current Instruction Register (CIR) Memory Address Register (MAR) Program Counter (PC) Accumulator (ACC) Index Register (IX) Status Register Interrupt Register	2
----------	---	----------

9608_s16_qp_12_sc_3.a

QUESTION

- 3 (a)** Describe how special purpose registers are used in the fetch stage of the fetch-execute cycle.

.....

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

3 (a) Four points from: - The Program Counter (PC) holds the <u>address</u> of the next <u>instruction</u> to be fetched - The address in the Program Counter (PC) is copied to the Memory Address Register (MAR) - The Program Counter (PC) is incremented - The instruction is copied to the Memory Data Register (MDR) from the <u>address</u> held in the Memory Address Register (MAR) - The instruction from the Memory Data Register (MDR) is copied to the Current Instruction Register (CIR)	[4]
--	------------

QUESTION

- 3 (a) Describe how special purpose registers are used in the fetch stage of the fetch-execute cycle.

.....

.....

.....

.....

.....

.....

.....

.....[4]

MARK SCHEME

- 3 (a) Four points from: [4]

- The Program Counter (PC) holds the address of the next instruction to be fetched
- The address in the Program Counter (PC) is copied to the Memory Address Register (MAR)
- The Program Counter (PC) is incremented
- The instruction is copied to the Memory Data Register (MDR)
 - from the address held in the Memory Address Register (MAR)
- The instruction from the Memory Data Register (MDR) is copied to the Current Instruction Register (CIR)

9608_s15_qp_13_sc_5.b.i

QUESTION

- (b) The sequence of operations shows, in register transfer notation, the fetch stage of the fetch-execute cycle.

```
1  MAR ← [PC]
2  PC  ← [PC] + 1
3  MDR ← [[MAR]]
4  CIR ← [MDR]
```

- [register] denotes contents of the specified register or memory location
- step 1 above is read as “the contents of the Program Counter are copied to the Memory Address Register”

- (i) Describe what is happening at step 2.

.....
..... [1]

MARK SCHEME

(b) (i) the program counter is incremented [1]

9608_s15_qp_13_sc_5.b.ii

QUESTION

- (ii) Describe what is happening at step 3.

.....
.....
..... [1]

MARK SCHEME

(ii) the data stored at the address held in MAR is copied into the MDR [1]

9608_s15_qp_13_sc_5.b.iii

QUESTION

- (iii) Describe what is happening at step 4.

.....
..... [1]

MARK SCHEME

- (iii) the contents of the Memory Data Register is copied into the Current Instruction Register

[1]

9608_s15_qp_13_sc_5.c

QUESTION

- (c) Describe what happens to the registers when the following instruction is executed:

LDD 35

.....
.....
.....
..... [2]

MARK SCHEME

- (c) • the MAR is loaded with the operand of the instruction // loaded with 35
• the Accumulator is loaded with the contents of the address held in MAR
// the Accumulator is loaded with the contents of the address 35

[2]

QUESTION

(c) The table shows six stages in the von Neumann fetch-execute cycle.

Put the stages into the correct sequence by writing the numbers 1 to 6 in the right hand column.

Description of stage	Sequence number
the instruction is copied from the Memory Data Register (MDR) and placed in the Current Instruction Register (CIR)	
the instruction is executed	
the instruction is decoded	
the address contained in the Program Counter (PC) is copied to the Memory Address Register (MAR)	
the value in the Program Counter (PC) is incremented so that it points to the next instruction to be fetched	
the instruction is copied from the memory location contained in the Memory Address Register (MAR) and is placed in the Memory Data Register (MDR)	

[6]

MARK SCHEME**(c)**

Description of stage	Sequence number
the instruction is copied from the Memory Data Register (MDR) and placed in the Current Instruction Register (CIR)	3
the instruction is executed	6
the instruction is decoded	5
the address contained in the Program Counter (PC) is copied to the Memory Address Register (MAR)	1
the value in the Program Counter (PC) is incremented so that it points to the next instruction to be fetched	4
the instruction is copied from the memory location contained in the Memory Address Register (MAR) and is placed in the Memory Data Register (MDR)	2

[6]

QUESTION

(c) The table shows six stages in the von Neumann fetch-execute cycle.

Put the stages into the correct sequence by writing the numbers 1 to 6 in the right hand column.

Description of stage	Sequence number
the instruction is copied from the Memory Data Register (MDR) and placed in the Current Instruction Register (CIR)	
the instruction is executed	
the instruction is decoded	
the address contained in the Program Counter (PC) is copied to the Memory Address Register (MAR)	
the value in the Program Counter (PC) is incremented so that it points to the next instruction to be fetched	
the instruction is copied from the memory location contained in the Memory Address Register (MAR) and is placed in the Memory Data Register (MDR)	

[6]

MARK SCHEME**(c)**

Description of stage	Sequence number
the instruction is copied from the Memory Data Register (MDR) and placed in the Current Instruction Register (CIR)	3
the instruction is executed	6
the instruction is decoded	5
the address contained in the Program Counter (PC) is copied to the Memory Address Register (MAR)	1
the value in the Program Counter (PC) is incremented so that it points to the next instruction to be fetched	4
the instruction is copied from the memory location contained in the Memory Address Register (MAR) and is placed in the Memory Data Register (MDR)	2

[6]

Show understanding of the purpose of interrupts

9608_s19_qp_13_sc_4.c.i

QUESTION

(c) The processor handles interrupts within the fetch-execute cycle.

(i) Give **one** example of a hardware interrupt and **one** example of a software interrupt.

Hardware

.....

Software

.....

[2]

MARK SCHEME

4(c)(i)	1 mark for hardware interrupt 1 mark for software interrupt For example: Hardware interrupt Printer out of paper No CD in drive Software interrupt A running program needs input Runtime error, e.g. division by zero	2
---------	--	----------

QUESTION

(ii) Explain how the processor handles an interrupt.

.....

.....

.....

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.....

.....

.....

.....

..... [5]

MARK SCHEME

4(c)(ii)	1 mark per bullet to max 5	5
	At the start / end of each fetch-execute cycle the processor checks for interrupt(s) Check if an interrupt flag is set // Check if bit set in interrupt register Processor identifies source of interrupt Processor checks priority of interrupt If interrupt priority is high enough // Lower priority interrupts are disabled Processor saves current contents of registers // saves current job on stack Processor calls interrupt handler / Interrupt Service Routine (ISR) Address of ISR is loaded into Program Counter (PC) When servicing of interrupt complete, processor restores registers // job from stack is restored Lower priority interrupts are re-enabled Processor continues with next F–E cycle	

QUESTION

(b) One stage of the FE cycle includes checking for interrupts.

State what is meant by an **interrupt**.

.....

.....

.....

.....[2]

MARK SCHEME

6(b)	1 mark per bullet point to max 2 • A signal from a source / device • Telling the processor its attention is needed	2
------	--	----------

QUESTION

(b) One stage of the FE cycle includes checking for interrupts.

(i) Give **three** different events that can generate an interrupt.

1

2

3 [3]

MARK SCHEME

4(b)(i)	1 mark for each event to max 3 For example: • Hardware fault // Example of hardware fault • I/O request // Example of I/O request • Program/software error // Example of software error • End of a time-slice	3
---------	--	----------

QUESTION

(ii) Explain how interrupts are handled during the fetch-execute cycle.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

MARK SCHEME

4(b)(ii)	1 mark per bullet point to max 5 • At the <u>end</u> of each fetch–execute cycle the processor checks for interrupt(s) • Check if an interrupt flag is set // Check if bit set in interrupt register • Processor identifies source of interrupt • Processor checks priority of interrupt • If interrupt priority is high enough // Lower priority interrupts are disabled • Processor saves current contents of registers • Processor calls interrupt handler / Interrupt Service Routine (ISR) • Address of ISR is loaded into Program Counter (PC) • When servicing of interrupt complete, processor restores registers • Lower priority interrupts are re-enabled • Processor continues with next F–E cycle	5
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QUESTION

- (b) Use the statements A, B, C and D to complete the description of how the fetch-execute cycle handles an interrupt.

A	the address of the Interrupt Service Routine (ISR) is loaded to the Program Counter (PC).
B	the processor checks if there is an interrupt.
C	when the ISR completes, the processor restores the register contents.
D	the register contents are saved.

At the end of the cycle for the current instruction

If the interrupt flag is set,, and

The interrupted program continues its execution.

[4]

MARK SCHEME

- (b) **One mark** for each statement or letter in the correct place.

[4]

At the end of the cycle for the current instruction **B**

If the interrupt flag is set, **D, A** and **C**

The interrupted program continues its execution

At the end of the cycle for the current instruction the processor checks if there is an interrupt.

If the interrupt flag is set, the register contents are saved, the address of the Interrupt Service Routine (ISR) is loaded to the Program Counter (PC) and when the ISR completes, the processor restores the register contents.

The interrupted program continues its execution.

QUESTION

- (b) Use the statements A, B, C and D to complete the description of how the fetch-execute cycle handles an interrupt.

A	the address of the Interrupt Service Routine (ISR) is loaded to the Program Counter (PC).
B	the processor checks if there is an interrupt.
C	when the ISR completes, the processor restores the register contents.
D	the register contents are saved.

At the end of the cycle for the current instruction

If the interrupt flag is set,, and

The interrupted program continues its execution.

[4]

MARK SCHEME

- (b) **One mark** for each statement or letter in the correct place.

[4]

At the end of the cycle for the current instruction **B**

If the interrupt flag is set, **D, A** and **C**

The interrupted program continues its execution

At the end of the cycle for the current instruction the processor checks if there is an interrupt.

If the interrupt flag is set, the register contents are saved, the address of the Interrupt Service Routine (ISR) is loaded to the Program Counter (PC) and when the ISR completes, the processor restores the register contents.

The interrupted program continues its execution.

QUESTION

- (d) (i) Explain what is meant by an interrupt.

.....

 [2]

MARK SCHEME

- (d) (i) • a signal
• from a device/program that it requires attention from the processor

[2]

9608_s15_qp_13_sc_5.d.ii

QUESTION

- (ii) Explain the actions of the processor when an interrupt is detected.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

- (ii) • at a point during the fetch-execute cycle ...
• check for interrupt
• if an interrupt flag is set/ bit set in interrupt register
• all contents of registers are saved
• PC loaded with address of interrupt service routine

[4]